Frequently Asked Questions

Answers to common questions



Unanswered Questions

If you have a question that is not answered here, please let me know by sending me an e-mail to tobias.vlcek@uni-hamburg.de.

Contribution and Mistakes

If you have found a mistake in the course material or if you have any suggestion on how to improve the course, please let me know by sending me an e-mail to tobias.vlcek@uni-hamburg.de.

FAQs

Is the course difficult?

In terms of difficulty, this is of course somehow complicated for me to assess. However, I have tried to make all the content as comprehensible as possible and really do ask you many questions in the lectures. If I have the impression that something has not been understood at all, then I take longer at that point. This semester is the first time I think the course is truly finished. I have adapted the slides based on feedback from the last two years and made all the examples as simple as possible. **In a nutshell**: If you are interested in the subject and equations do not scare you, you will likely do great in the course.

How can I download PDF slides from the lecture?

Open the lecture and click on **Slides (PDF)** in the top right corner: the slides are there as a finished PDF to download. The **PDF** button next to it gives you the lecture as a text document instead.

How does the calculation of the bonus points work?

Each assignment gives up to 5 points in Moodle, depending on how much of it you solved correctly. Divided by 10, that is up to 0.5 bonus points per assignment for the exam. In total, you can earn at most 6.0 bonus points, and they only count if the exam is passed.

How do the groups for the assignments work?

You hand in the assignments as a group of up to three; working alone is fine too. The groups are formed in the second lecture and stay the same for the whole semester. If you missed it or have no group yet, send me a short email.

Are the solutions to the tutorials published?

No. Instead, I give you feedback on your group's submission.

Can I take a sheet with me to the exam?

Yes, you can take a DIN A4 sheet with notes on both sides to the exam. Please note, that it should be **one** sheet of paper.

How does the chatbot work?

The course AI is a chatbot that uses the Mistral and some custom code hosted on Hetzner in Germany. If you have any questions about the course, feel free to ask the AI. Note, that the AI is not perfect and sometimes the answers might be incorrect. For more information about how the data is processed, please refer to the privacy policy.

My Julia code does not run in Zed. What can I do?

The commands in **bold** are in Zed's command palette (`Ctrl+Shift+P`, macOS `Cmd+Shift+P`).

- **Nothing happens when I press `Ctrl+Shift+Enter`.** Save the file with the ending `.jl` and press the shortcut a second time: right after Zed starts, the first press can get lost while Zed looks for its kernels. Each cell has to start with `# %%` at the very beginning of a line.
- **Zed does not list the kernel `applied-optimization-1.13`.** Run **repl: refresh kernelspecs**. If it is still missing, repeat steps 3 to 5 of the first tutorial and look for error messages after `installkernel`.
- **`ArgumentError: Package JuMP not found in current path`** (or any other package). Zed started another kernel, for example `julia-1.13`, which does not use the course environment. Check the kernel selector at the top right of the editor and choose `applied-optimization-1.13`. If the kernel is right, the package is not installed yet: run `using Pkg; Pkg.add("JuMP")` in a cell.
- **`UndefVarError: ... not defined`.** The cell uses a variable from a cell that has not run yet. Run the cells above it first. After **repl: restart**, all variables are gone and you start again from the top.

- **Plots do not appear.** Your kernel was installed without the `zed_plots` part of step 5 in the first tutorial: close Zed and run steps 3 and 5 again. Interactive plots made with `plotly()` cannot be shown in Zed; leave that line out.
- **A model keeps solving and does not stop. repl: interrupt** cannot stop a running solver. Use **repl: restart** instead and run your cells again from the top.
- **The kernel does not start after a Julia update.** The kernel still points to the old Julia version, which the update removed. Close Zed and run steps 3 and 5 of the first tutorial again.
- **The first cell takes minutes.** Julia compiles each package once on your computer, so the first `using JuMP` or `using Plots` is slow, on Windows even slower because Microsoft Defender scans the new files. Let it finish: afterwards, the same commands are fast.
- **An exclamation mark appears in Zed's title bar, and code completion does not work.** Zed opened your folder in Restricted Mode. Click the exclamation mark and trust your course folder (only that folder, not your whole user folder).
- **`Ctrl+C` does not copy in Zed's terminal (Windows, Linux).** In the terminal, `Ctrl+C` stops the running program; copy with `Ctrl+Shift+C`.

If Zed itself does not run on your computer, use the VS Code fallback described in the first tutorial.

Julia or Zed does not work on my Windows laptop. What can I do?

- **The terminal does not find `julia`.** Open a new terminal; if you use the terminal inside Zed, restart Zed. If that does not help, open *Settings → Apps → Advanced app settings → App execution aliases* and switch on `julia.exe` and `juliaup.exe`. If the Microsoft Store is blocked on your laptop, use the MSI installer instead.
- **"Program 'julia.exe' failed to run: The file cannot be accessed by the system".** An installation through the Microsoft Store, `winget` or the MSIX App Installer did not finish and left a broken alias for `julia.exe` behind. The same alias makes `;julia` inside Julia fail with `permission denied (EACCES)`. Uninstall *Julia* under *Settings → Apps → Installed apps*. If you also installed a standalone version such as the 1.10 LTS, uninstall it too: the course runs Julia 1.13 through juliaup. If `julia` still shows the error, switch off `julia.exe` and `juliaup.exe` under *Settings → Apps → Advanced app settings → App execution aliases*. Then install Julia with the MSI installer and open a new terminal; if you use the terminal inside Zed, restart Zed. If you had already installed the course kernel, run steps 3 and 5 of the first tutorial again.
- **"An Application Control policy has blocked this file", "Smart App Control has blocked part of this app" or error `0xc0e90002`.** Windows Smart App Control blocks the package files that Julia compiles on your laptop. Switch it off under *Windows Security → App & browser control → Smart App Control* and keep it off during the course: switched on again, it can block every package Julia compiles later. Then delete the folder `%USERPROFILE%\.julia\compiled\v1.13` and run `using Pkg;`

`Pkg.precompile()` in the course environment. On a laptop managed by your university or employer, only the IT department can allow these files; until then, start Julia with `julia --pkgimages=no --project=applied-optimization` in the first tutorial and add `"--pkgimages=no"` right after the `--project` argument of its `installkernel` call. Packages then load more slowly, but they load.

- **Precompiling IJulia fails with an `EPIPE` error.** Run `using IJulia` again: the error is known on Windows and usually disappears on the second attempt.
- **The Git panel in Zed reports "dubious ownership" or cannot initialize a repository.** Zed needs Git itself: install Git for Windows and restart Zed.
- **Your laptop has a Snapdragon (ARM) processor.** Julia runs there in x64 emulation: the course setup should work the same way, only slower.
- **Your course folder lies in OneDrive.** Synced folders can block Julia from replacing files. Use a plain folder such as `C:\Users\<name>\ao`.

Still stuck? Bring your laptop to the tutorial and we'll fix it together.

Bibliography